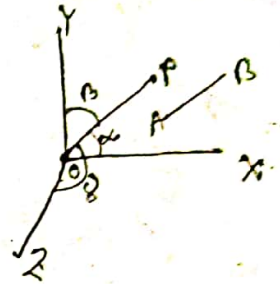
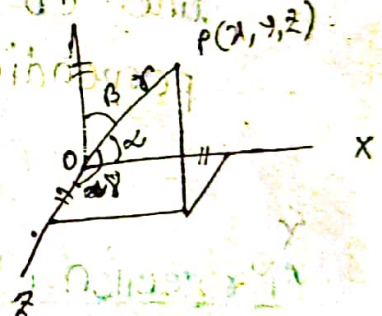


* Direction Cosines of a line:

If a given line AB makes angle α, β, γ with the positive direction of axis x, y, z respectively. Then $\cos \alpha, \cos \beta, \cos \gamma$ are called the direction-cosines of the line and it is denoted by the letters l, m, n respectively.



Ex Prove that, if l, m, n be the DC of any line, then $l^2 + m^2 + n^2 = 1$.



Proof: Let $P(x, y, z)$

Again let $OP = r$ and the direction angles

of $OP = [\alpha, \beta, \gamma]$

so that $OP = r = \sqrt{x^2 + y^2 + z^2}$

and the DC of $OP = [\cos \alpha, \cos \beta, \cos \gamma] = [l, m, n]$

From the diagram we have,

$$x = OP \cos \alpha = rl \quad \text{--- (i)}$$

$$y = OP \cos \beta = rm \quad \text{--- (ii)}$$

$$z = OP \cos \gamma = rn \quad \text{--- (iii)}$$

From (i), (ii) & (iii) we have

$$x^2 + y^2 + z^2 = r^2 (l^2 + m^2 + n^2) = r^2$$

$$\Rightarrow r^2 = r^2 (l^2 + m^2 + n^2)$$

$$\therefore l^2 + m^2 + n^2 = 1. \text{ Hence proved.}$$

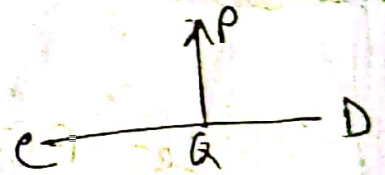
* Direction ratio of a line:

Any three numbers a, b, c which are not zero and proportional to the direction cosines l, m, n respectively of a given line is called direction ratios of the given line.

Now from the definition of D.R. -

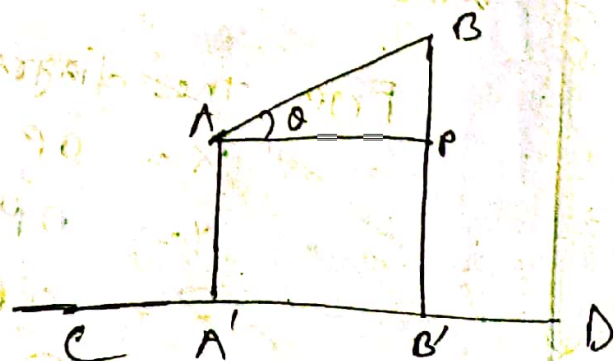
$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{a^2 + b^2 + c^2}} \\ = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}}.$$

* Projection of a pt: The projection of a pt. P on a given line CD is the pt. Q which is the foot of the perpendicular from P on CD .



* Projection of a line: The projection of a given line AB on another line CD is the segment $A'B'$ of CD where A' and B' are the projection of the points A and B on the line CD .

$$A'B' = AB \cos \theta.$$



Find, the DC of the line joining given pt. $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$.

Solⁿ

Let, the direction angles of the line $PQ = [\alpha, \beta, \gamma]$
So that, the DC of the line $PQ = [\cos \alpha, \cos \beta, \cos \gamma]$
= $[l, m, n]$

Here, $OP' = x_1$ & $OQ' = x_2$

$$\therefore P'Q' = x_2 - x_1$$

We know,

$$P'Q' = PQ \cos \alpha$$

$$\Rightarrow \cos \alpha = \frac{P'Q'}{PQ}$$

$$\therefore l = \frac{x_2 - x_1}{PQ}$$

Similarly, $m = \frac{y_2 - y_1}{PQ}$ and $n = \frac{z_2 - z_1}{PQ}$

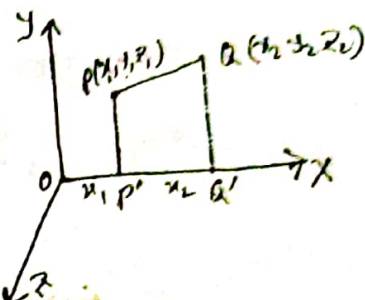
PQ is the DC of l, m, n

$$\therefore PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$\therefore \cos \alpha = \frac{x_2 - x_1}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

$$\cos \beta = \frac{y_2 - y_1}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

$$\& \cos \gamma = \frac{z_2 - z_1}{\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$



Q Find D.C of the line by joining given two pt. $(5, 6, 7)$ & $(4, 10, -1)$.

Soln:

The distance between two pt. $(5, 6, 7)$ & $(4, 10, -1)$

$$\text{is, } = \sqrt{(5-4)^2 + (6-10)^2 + (7+1)^2}$$

$$= \sqrt{1+16+64}$$

$$= \sqrt{81}$$

$$= 9$$

Suppose that, the line makes angle α , β & γ with the axis OX , OY and OZ respectively.

$$\therefore \cos \alpha = \frac{4-5}{9} = -\frac{1}{9}$$

$$\cos \beta = \frac{10-6}{9} = \frac{4}{9}$$

$$\& \cos \gamma = \frac{-1-7}{9} = -\frac{8}{9}$$

Ex. If the coordinates of the pt. P, Q, R, S are $(3, 4, 5)$, $(4, 6, 3)$, $(-1, 2, 4)$, $(1, 0, 5)$. Then find the DC of PQ and RS. Also find the projection of RS on PQ.

Soln Here, $PQ = \sqrt{(3-4)^2 + (4-6)^2 + (5-3)^2} = \sqrt{1+4+4} = 3$
 $RS = \sqrt{(-1-1)^2 + (2-0)^2 + (4-5)^2} = \sqrt{9} = 3$

DC of PQ are, $\frac{4-3}{3}, \frac{6-4}{3}, \frac{3-5}{3}$

$= \frac{1}{3}, \frac{2}{3}, -\frac{2}{3}$.

Again, DC of RS are, $\frac{1+1}{3}, \frac{0-2}{3}, \frac{5-4}{3}$
 $= \frac{2}{3}, -\frac{2}{3}, \frac{1}{3}$.

Now projection of RS on PQ is,

$$(1+1)\frac{1}{3} + (0-2)\frac{2}{3} + (5-4)(-\frac{2}{3})$$

$$= \frac{2}{3} - \frac{4}{3} - \frac{2}{3}$$

$$= -\frac{4}{3}$$

